

MANSOOR MAMNOON

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EDUCATION

University of California - Berkeley

Bachelor's, Computer Science

Operating Systems, Compilers, Database Systems, Efficient Algorithms, Data Structures and Algorithms, Computer Architecture, Machine Learning, Optimization Methods, Probability and Statistics, Linear Algebra, Discrete Mathematics, Honors Abstract Linear Algebra, Computer Vision

Aug 2023 - May 2027

GPA: 3.973

PROFESSIONAL EXPERIENCE

Databricks

Software Engineer Intern

- Developing dynamic network interface attachment for GCP-provisioned VMs, enabling transparent enforcement of customer network isolation guarantees across multi-tenant cloud infrastructure
- Implementing cloud provider integrations within an internal network control plane, interfacing with service network layers to maintain workload boundary semantics at scale
- Designed and validated a proof-of-concept by manually provisioning and configuring VMs end-to-end, confirming architectural feasibility before full implementation
- Collaborating with senior infrastructure engineers on architecture design, presenting implementation plans for review prior to development

May 2026 - Present

Bellevue, WA, USA

Berkeley Artificial Intelligence Research

Undergraduate Researcher – AUTOLab (BAIR)

- Working with postdoctoral researchers in UC Berkeley's AUTOLab (affiliated with Berkeley AI Research, BAIR) on extensions to Robo-DM, a data management toolkit for large-scale robot learning datasets.
- Exploring efficient dataset compression, retrieval, and integration with ML training pipelines.
- Applying prior experience in real-time pipelines (Amazon SDE intern) to robotics data loading and systems optimization.

Sep 2025 - Dec 2025

Berkeley, CA, USA

Amazon

Software Development Engineer Intern

- Architected and shipped a real-time occupancy monitoring platform (100+ IoT sensors, 10K+ events/min, P99 latency <500ms), piloted across 4 Amazon buildings with expansion underway.
- Built a scalable, fault-tolerant backend on AWS (ECS Fargate, SQS, DynamoDB, Timestream), cutting infra costs and enabling horizontal scaling under peak loads.
- Automated multi-region deployments via AWS CDK (TypeScript), reducing provisioning time by 85% and standardizing infrastructure across dev/staging/prod.
- Engineered a React dashboard with <1s data freshness, enabling ops teams to monitor usage live and improve space allocation across facilities.
- Authored design doc + exec one-pager, led review cycles with SDE2/SDE3s, and incorporated feedback into final deployed architecture.

May 2025 - Aug 2025

Seattle, WA, USA

PROJECTS & OUTSIDE EXPERIENCE

Order Book Engine – High-Performance Trading Infrastructure

Software Engineer (Independent Project)

- Built a C++20 matching engine with slab allocators, intrusive FIFO queues, side-specialized templates for branch elimination, and cache-hot best-level pointers, achieving 20M+ msgs/sec throughput and p50=0.04µs latency; validated determinism via snapshot/resume replay proof with bit-for-bit fill equivalence across 3.5M+ real BTCUSDT events from Binance US
- Designed an end-to-end analytics pipeline (Python, Parquet, Docker, Streamlit) computing microstructure metrics — spread, microprice, realized volatility (Parkinson & Garman-Klass), price impact curves, and order-flow autocorrelation — from live WebSocket market data replayed through the C++ engine
- Implemented and benchmarked four execution strategies (TWAP, VWAP, POV, Iceberg) with a full cost model (fees, latency, tick rounding), PnL/risk metrics (Sharpe, max drawdown, CVaR), and deterministic reproducibility via SHA-256 checksums across parallel parameter sweeps
- Shipped a containerized analytics pipeline (Docker, GHCR) with GitHub Actions CI/CD, generating self-contained HTML reports with embedded plots and JSON summaries from a single command; exposed C++ engine to Python via pybind11 bindings
- [Link to project](#)

Jul 2025 - Aug 2025

Berkeley, CA, USA

LLMFirewall – AI Agent Security

Software Engineer (Independent Project)

- Built an MCP security proxy enforcing taint-aware least privilege for tool-using LLM agents, intercepting every tool call and resource read across 7 independent enforcement layers at sub-millisecond latency with no external model calls
- Designed an information-flow control system with source-level taint labels, categorically preventing untrusted retrieved content from authorizing write-side-effect tools regardless of detection score: reducing attack success rate from 73% to 0% on capability-exploiting attacks
- Evaluated across 4,200+ cases spanning direct injection, indirect RAG, multi-turn escalation, and hard benign negatives, achieving 99.4% precision and 0% false positive rate among scoring-based defenses
- Validated against real MCP servers (filesystem, git, SQLite) across 26 integration scenarios; shipped as installable Python package with Claude Desktop and Cursor integration guides
- [Link to project](#)

Dec 2025 - May 2026

Berkeley, CA, USA

Edge Deployer – Serverless IDE for Multi-Cloud Deployment

Software Engineer (Independent Project)

- Engineered a native desktop IDE (Electron, React 19, TypeScript) unifying the full serverless development loop — Monaco editor, live edge runtime simulator, one-click multi-cloud deployment to Cloudflare Workers, AWS Lambda, Vercel, Netlify, [Fly.io](#), and Railway — across 6 providers via a shared 7-method deployer interface
- Built a sandboxed plugin system using Node.js vm.Context with explicit permission manifests, enforcing capability isolation (no filesystem/network access unless declared) with 5-second execution timeouts and hook-based lifecycle integration across deploy, rollback, and code transform stages
- Implemented an AES-256-GCM encrypted secrets vault with PBKDF2 key derivation (100,000 iterations), structurally excluding credentials from workspace files at the TypeScript type level to make accidental secret commits impossible
- Shipped cross-platform binaries (macOS universal, Windows, Linux) via GitHub Actions CI/CD with matrix builds, 85-test suite across 6 suites covering security scanner, load test stats, deployer compliance, plugin sandbox, vault encryption, and workspace persistence
- [Link to project](#)

Mar 2025 - May 2025

Berkeley, CA, USA